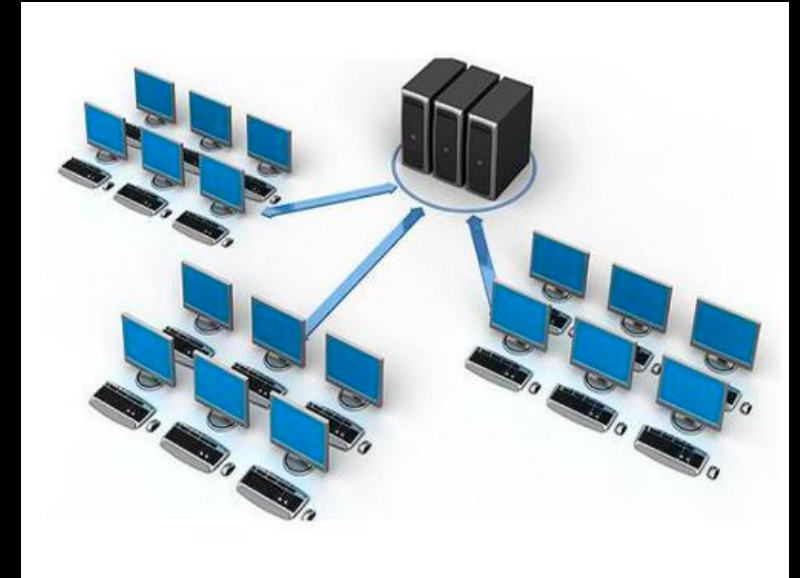




Networking

What is network ?

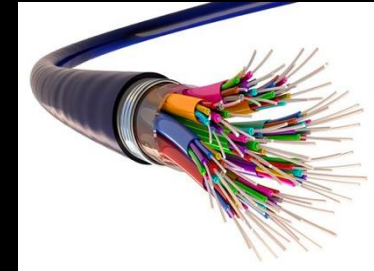
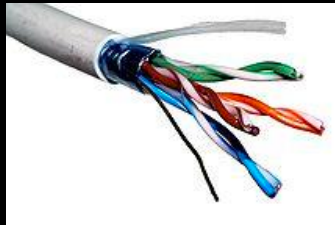
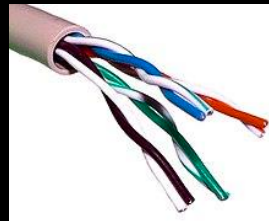
- It is the interconnection of multiple devices, generally termed as Hosts connected using multiple paths
- The purpose of network is
 - sending/receiving data or media
- It involves various devices like hubs, switches, routers etc.



Wired network



- The network build by connecting devices together using wires/cables as a medium to transfer the data
- Cables
 - Coaxial cable
 - Twisted pairs cables
 - Fiber optics



Wireless network

- The network build by connecting devices together using air as a medium to transfer the data
- EM Waves are used to transfer data from sender to receiver



Network Types



- **Personal Area Network**
 - Smallest network which is very personal to the user
 - E.g. BlueTooth
- **Local Area Network**
 - Spans across building(s) and operated under single administrative system
 - E.g. company, school network
 - Technologies: TokenRing or Ethernet
- **Metropolitan Area Network**
 - Spans across cities
 - E.g. cable network
 - Technologies: high speed fiber optics
- **Wide Area Network**
 - Spans across countries
 - Technologies: ATM, Frame Relay



What is a network topology ?

- Physical arrangement of computers is known as topology
- Famous topologies
 - Bus
 - Ring
 - Token Ring
 - Star
 - Mesh

ISO OSI model



- Conceptual model that characterizes and standardizes the communication functions of a telecommunication or computing system without regard to its underlying internal structure and technology
- Goal is the interoperability of diverse communication systems with standard communication protocols
- Layered architecture having 7 layers
 - Application
 - Presentation
 - Session
 - Transport
 - Network
 - Data Link
 - Physical

Application Layer



- Specifies interface methods used by hosts in a communications network
- Contains communication protocols
 - **HTTP [80]**: Hyper Text Transfer Protocol
 - **HTTPS [443]**: Secure Hyper Text Transfer Protocol
 - **FTP [20, 21]**: File Transfer Protocol
 - **SFTP [115]**: Simple FTP
 - **DNS [53]**: Domain Name Service
 - **NFS [1023]**: Network File System
 - **POP3 [110]**: Post Office Protocol
 - **SMTP [25]**: Simple Mail Transfer Protocol
 - **SSH [22]**: Secure Shell
 - **LDAP [389]**: Lightweight Directory Access Protocol

Presentation Layer



- Serves as the data translator for the network
- Also known as syntax layer
- Responsible for
 - Translation
 - Compression/Decompression
 - Encoding/Decoding
 - Encryption/Decryption

Session Layer

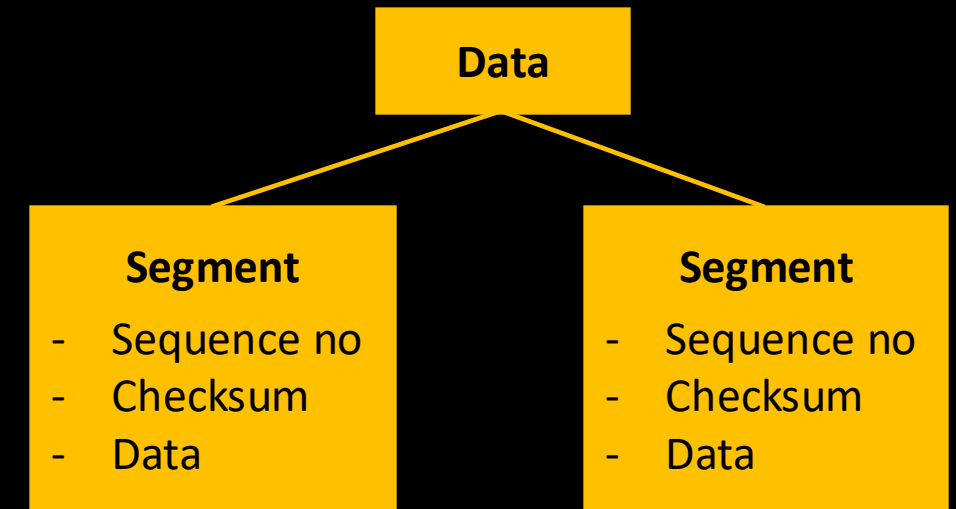


- Provides mechanism for opening, closing and managing session between processes
- Communication sessions consist of requests and responses that occur between applications
- Protocols
 - **ASP**: AppleTalk Session Protocol
 - **ADSP**: AppleTalk Data Stream Protocol
 - **NetBIOS**: Network BIOS
 - **PAP**: Password Authentication Protocol
 - **PPTP**: Point to Point Tunnelling Protocol
 - **RPC**: Remote Procedure Call
 - **SCP**: Session Control Protocol
 - **SDP**: Socket Direct Protocol

Transport Layer



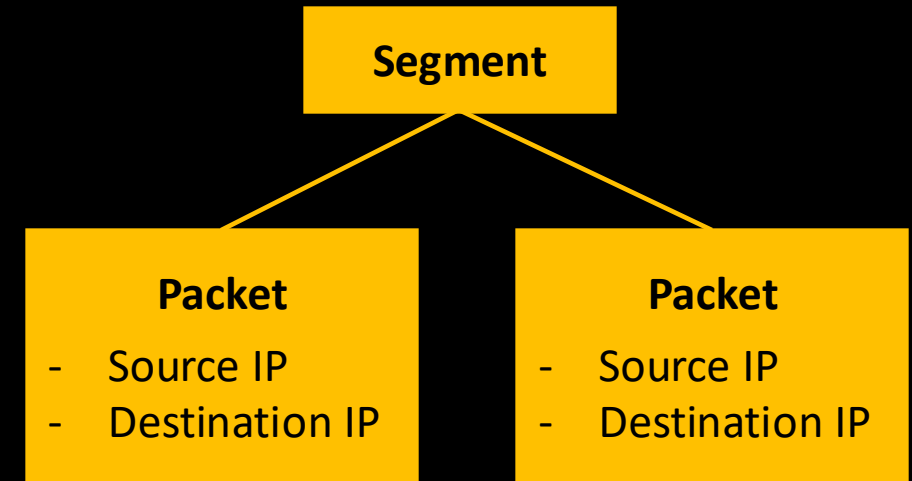
- Provide host-to-host communication services for applications
- Creates Segment (data unit) containing
 - Sequence number
 - Checksum
 - Port number
- Protocols
 - **TCP**
 - Connection oriented protocol
 - Provides: Flow Control, Error checking
 - Guarantees data delivery
 - Slower than UDP
 - E.g. WWW, HTTP
 - **UDP**
 - Connectionless protocol
 - Does not provide flow control
 - Does not guarantee data delivery
 - Faster than TCP
 - E.g. streaming, online games



Network Layer



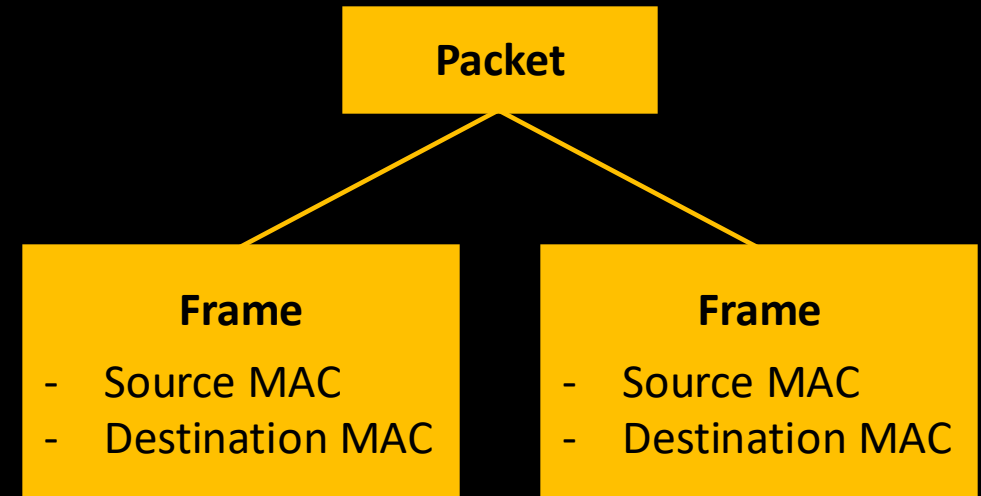
- Responsible for packet forwarding including routing through intermediate routers
- Responsible for splitting segment into packets containing
 - Source IP address
 - Destination IP address
- Protocols
 - **IP**: Internet Protocol
 - **IPX**: Internetwork Packet Exchange
 - **IPSec**: Internet Protocol Security
 - **EGP**: Exterior Gateway Protocol





Data Link Layer

- Transfers data between
 - adjacent network nodes in a wide area network (WAN) or
 - between nodes on the same local area network (LAN) segment
- Encapsulates packet into Frames containing
 - Source MAC Address
 - Destination MAC Address
- Sublayers
 - Logical Link Layer
 - Media Access Control Layer



Data Link Layer: Logical Link Layer



- The uppermost sublayer multiplexes protocols running at the top of data link layer, and optionally provides flow control, acknowledgment, and error notification
- Provides addressing and control of the data link
- Services
 - Error control (automatic repeat request, ARQ)
 - Flow control [Data-link-layer flow control is not used in LAN protocols such as Ethernet, but in modems and wireless networks]



Data Link Layer: Media Access Control Layer

- Refers to the sublayer that determines who is allowed to access the media at any one time (CSMA/CD)
- Determines where one frame of data ends and the next one starts (frame synchronization)
- Frame synchronization uses: time based, character counting, byte stuffing and bit stuffing.
- Services
 - Multiple access protocols for channel-access control,
 - CSMA/CD protocols for collision detection and re-transmission in Ethernet networks
 - CSMA/CA protocol for collision avoidance in wireless networks
 - Physical addressing (MAC addressing)
 - LAN switching (packet switching), including MAC filtering, Spanning Tree Protocol (STP) and Shortest Path Bridging (SPB)
 - Data packet queuing or scheduling



Physical Layer

- Consists of the electronic circuit transmission technologies of a network
- Fundamental layer underlying the higher level functions in a network which provides means of transmitting raw bits rather than logical packets or segments
- The bitstream may be grouped into code words or symbols and converted to a physical signal that is transmitted over a transmission medium
- Translates logical communications requests from the data link layer into hardware-specific operations to cause transmission or reception of electronic signals
- Services
 - Modulation/Demodulation
 - Multiplexing
- Consists of
 - Cables/wires
 - Devices like hub, repeaters etc.



Addressing Modes: MAC Address

- Used to identify NIC uniquely
- Consists of 6 bytes [48 bits]
- First 3 bytes represents manufacturer
- Next 3 bytes represents NIC's unique address



Addressing Modes: IP Address

- Used to identify every device uniquely
- Set by operating system running on the device
- Can be written in
 - Decimal: 192.168.100.10
 - Binary: 11000000.10101000.01100100.00001010
- Versions
 - **IPv4**
 - 32 bit [4 bytes] address
 - Classful and Classless addressing
 - **IPv6**
 - 128 bit address
- Types
 - Private: used to communicate with other devices in local network
 - Public: used to communicate with other devices over internet

environment → collection of components required to run/develop the app

→ dev env → developers
→ staging env → testers
→ pre-production env →
→ UAT env →
→ production env → end users

→ Resources

→ machines
→ Network
→ Software → frontend
→ backend
→ database
→ libraries/framework

Deployment

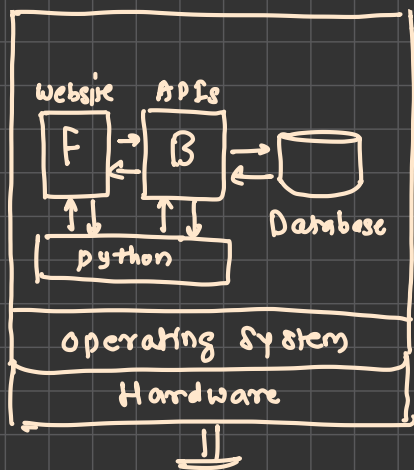
moving the app from one env to another

→ traditional
→ virtualized
→ containerized



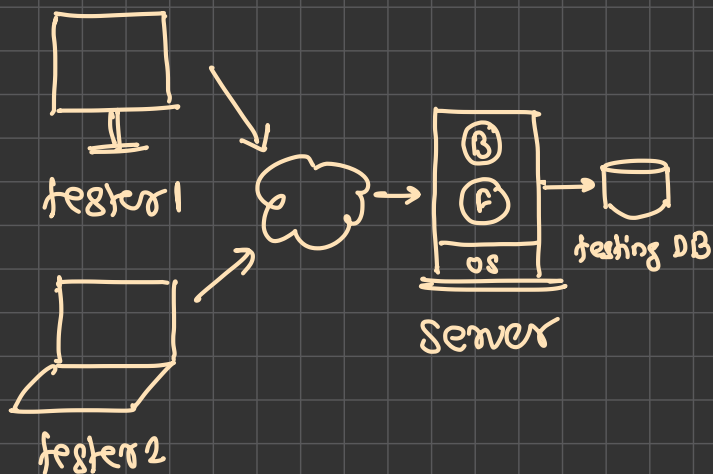
Dev environment

Developers



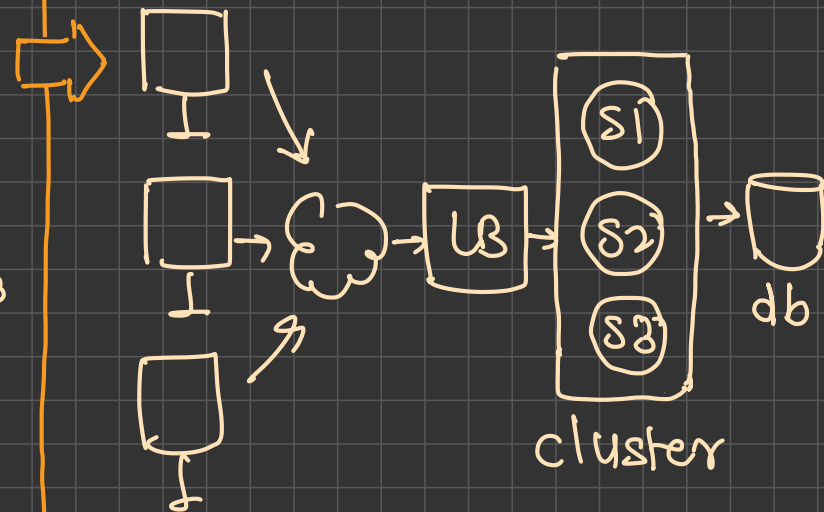
Staging / testing env

testers



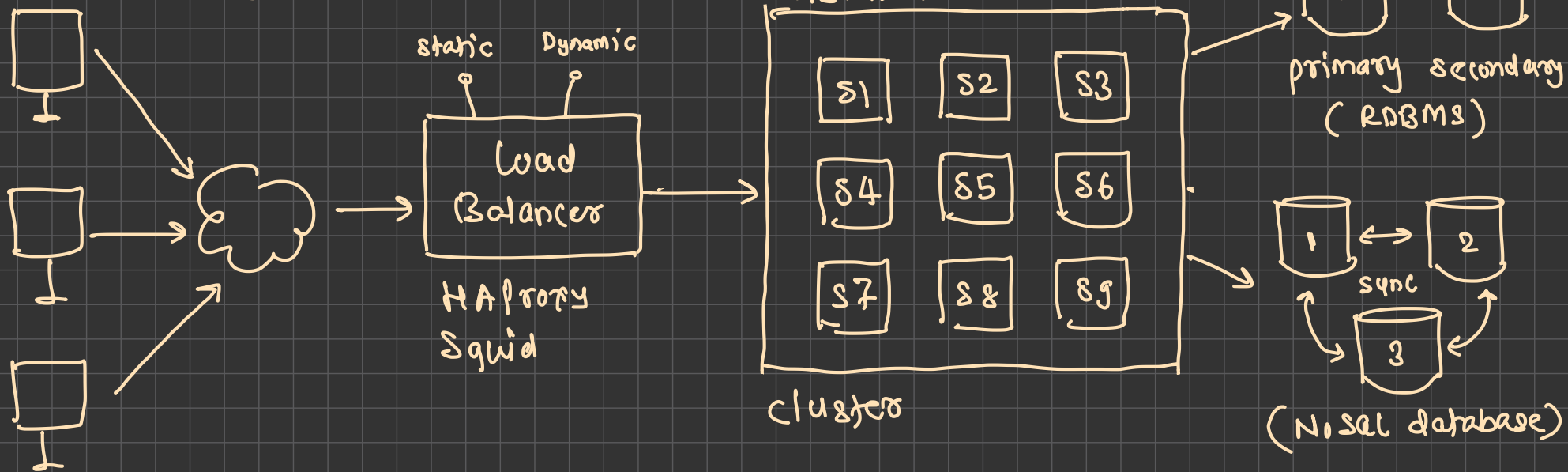
pre-production environment

used by testers → miniature of production env



live / production env → used by end users (using horizontal scaling)

↳ Highly Available env/app



Traditional Deployment → using physical machines



- Traditional deployment using physical machines (also called *on-premises deployment* or *on-prem*) refers to the old-style way of running applications before cloud computing existed
- It means you buy, own, and maintain all the hardware required to run your application:
 - Physical servers
 - Storage devices
 - Networking devices
 - Data center space
 - Power and cooling
 - IT staff
- Your application runs on machines inside your company's office or data center, not on the cloud

Expenditure

- Capital Exp (CapEx) ↑
 - upfront investment
 - purchasing hardware
 - mostly rare / one time action
- Operational Exp (OpEx)
 - maintenance cost
 - salaries paid to IT staff
 - recurring basis



How Traditional Deployment Works

- Buy Hardware → physical
 - Set Up the Data Center
 - Install the Operating System & Software
 - Network Configuration
 - Deploy the Application → app
 - Ongoing Maintenance
- database
→ dependencies
- libraries
frameworks
binaries

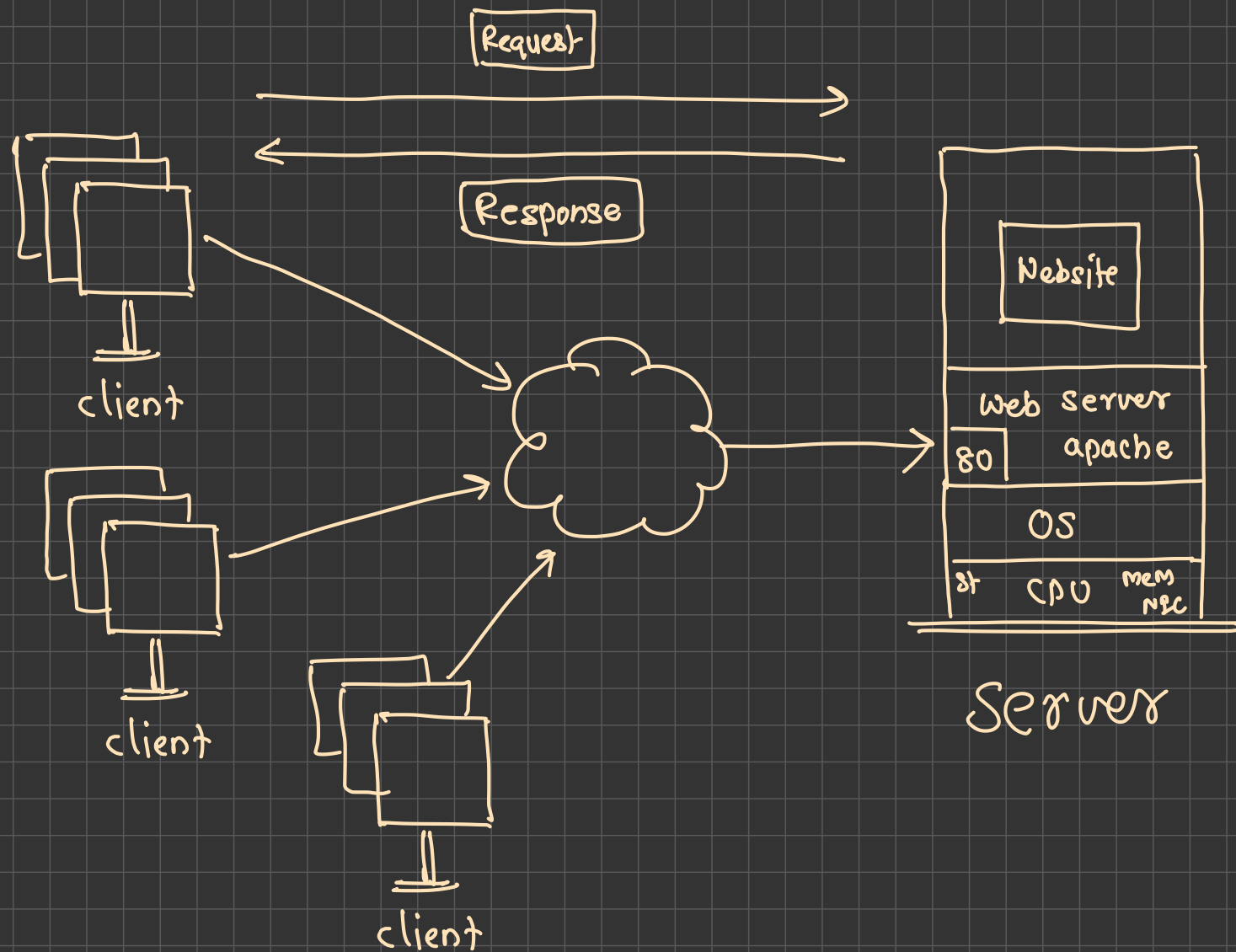


Limitations of Traditional Deployment

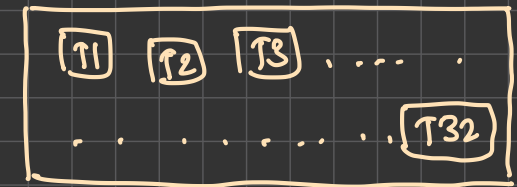
- No Elasticity / Hard to Scale
- High Upfront Cost → CapEx
- Slow Deployment → delivery / configuration
- Risky Failure Handling
- Limited Innovation Speed



Scaling



Thread Pool



server configuration

→ CPU ⇒ Xeon 3.4 GHz
→ 32 Threads

RAM ⇒ 1 TB

Storage = 150 TB

What is Scaling?



- **Scaling** means **increasing or decreasing the capacity of your system** (servers, computing power, storage, etc.) based on load or demand
- If your website normally gets **1,000 users/day**, but suddenly it gets **50,000 users/day**, you need more resources
- **Without scaling:**
 - **Website becomes slow**
 - **May crash**
 - **Users leave**
 - **Loss of business**
- **Scaling helps you handle traffic smoothly**
- **Types of scaling**
 - **Vertical Scaling**
 - **Horizontal Scaling**

Vertical Scaling

- Add or remove resources on machine

- E.g. add or remove resources
 - ~~Add more instances/servers~~
 - ~~Add a load balancer~~
 - ~~Share traffic across servers~~

- Advantages

- Simpler to Implement → requires less skills (HW knowledge)
- No Application Changes Needed
- Better for Monolithic Applications → one code base (git repo)
- Easier Maintenance
- Works Well for Databases

- Disadvantages

- Hardware Limitations → No infinite upgrade
- Downtime During Upgrades
- Single Point of Failure → less available
- Not Suitable for Massive Growth

→ CPU
→ storage
→ memory
→ network card
→ bandwidth

①

$$\text{CPU} = X_{\text{con}} \quad 3.4 \text{ GHz} \quad (32 \text{ T})$$

↓

$$\begin{aligned} \text{CPUs} &= 4 \times X_{\text{con}} \quad 3.4 \text{ GHz} \quad (32 \text{ T}) \\ &= 4 \times 32 = 128 \text{ T} \end{aligned}$$

②

$$\text{Memory} \Rightarrow 1 \text{ TB}$$

↓

$$2 \text{ TB}$$

Horizontal Scaling

- You add more servers instead of increasing the size of one server
- E.g.
 - Add more instances/servers
 - Add a load balancer
 - Share traffic across servers
- Advantages
 - Virtually Unlimited Scalability
 - Increased Reliability & High Availability (HA)
 - Zero or Minimal Downtime
 - Automatic Scaling ✖ ✖
 - Better Performance Under High Load
- Disadvantages
 - More Complex Architecture → more skills
 - More DevOps & Networking Knowledge Needed
 - Application Must Support It
 - Data Consistency Challenges
 - More Monitoring Needed
 - Cost Increases Exponentially

cluster → group of machines (servers)
known as nodes





Virtualization

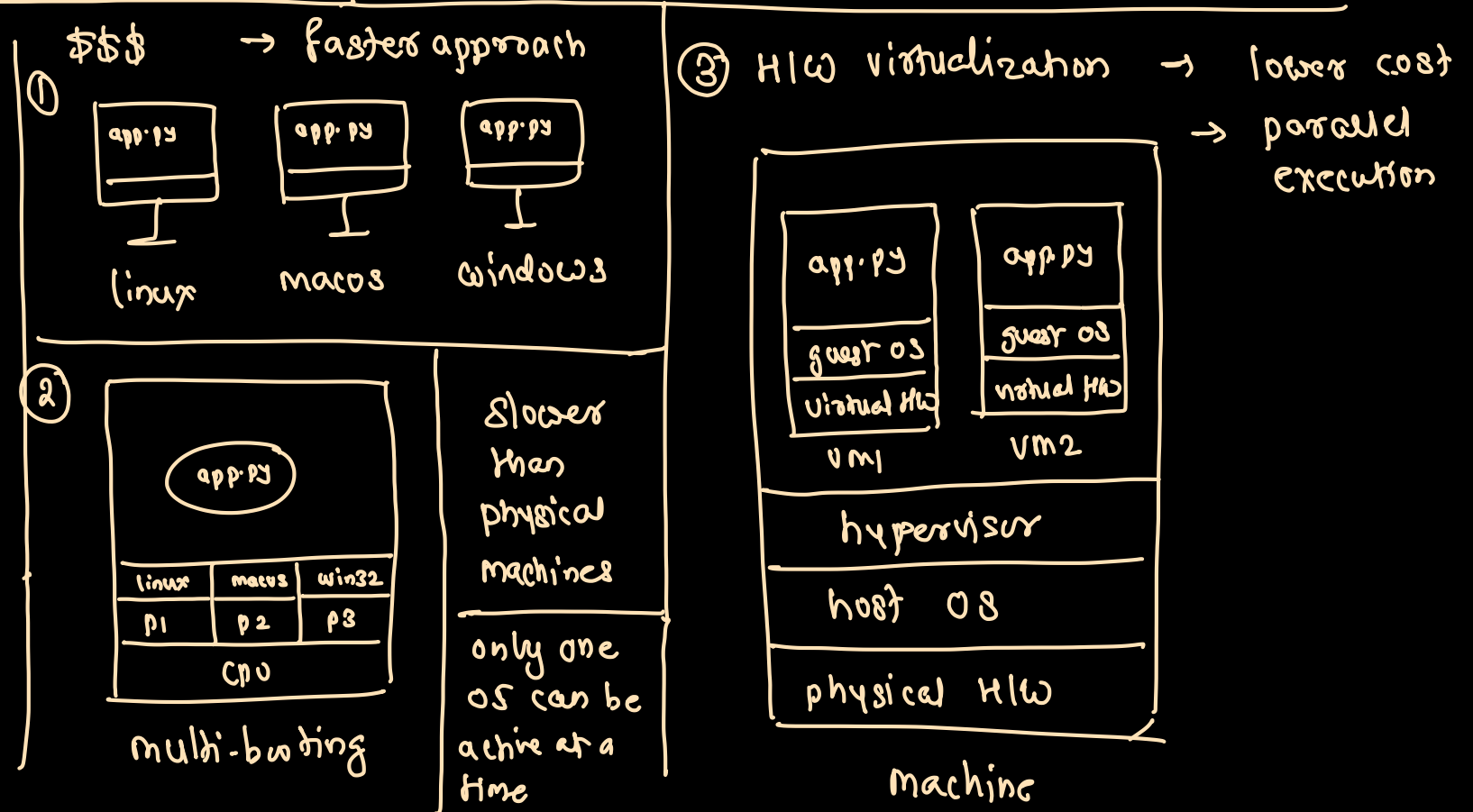
What is virtualization [sharing]



- Virtualization is the creation of a virtual -- rather than actual -- version of something, such as an operating system (OS), a server, a storage device or network resources
- Virtualization uses software that simulates hardware functionality in order to create a virtual system → hypervisor
- This practice allows IT organizations to operate multiple operating systems, more than one virtual system and various applications on a single server

Types

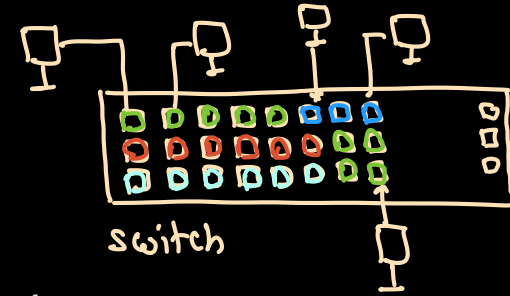
- Network virtualization
- Storage virtualization
- Data virtualization
- Desktop virtualization
- Application virtualization
- Hardware virtualization
- Operating System virtualization
(Containerization)



Network Virtualization



- Network virtualization is the process of abstracting and combining hardware and software network resources into a single, software-based administrative entity
- This approach allows multiple virtual networks to run on the same physical hardware while maintaining the flexibility to configure and manage them independently
- **Key Features of Network Virtualization**
 - **Abstraction:** It separates the physical network infrastructure from the logical network, enabling more flexibility
 - **Scalability:** Virtual networks can be scaled up or down without changing the underlying physical network
 - **Isolation:** Virtual networks can operate independently, ensuring that changes in one do not affect others
 - **Flexibility:** Administrators can design and manage virtual networks without being constrained by physical topology
- **Types of Network Virtualization**
 - **Internal Network Virtualization**
 - Consolidates multiple network components within a single physical network, allowing them to function as separate entities
 - Examples include VLANs (Virtual Local Area Networks) and virtual switches
 - **External Network Virtualization**
 - Combines multiple physical networks or network devices into a single virtualized unit
 - Examples include VPNs (Virtual Private Networks) and software-defined networking (SDN)
- **Use Cases**
 - **Cloud Computing:** Virtualized networks are fundamental to cloud platforms where resources are shared across tenants
 - **Data Centres:** Simplifies management and improves resource utilization in large-scale environments



Storage Virtualization → NAS / SAN

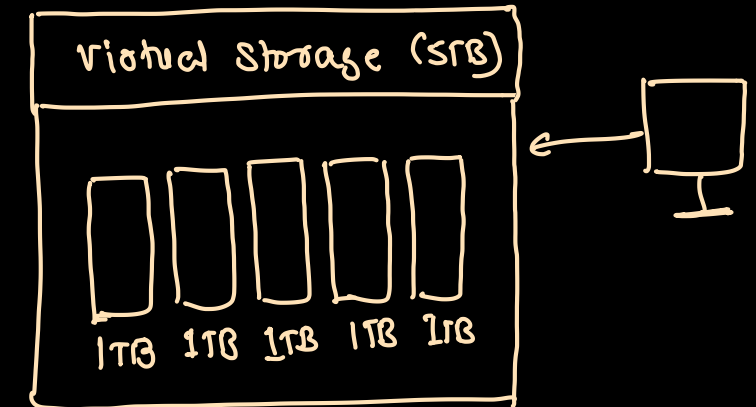


- Storage virtualization is the pooling of physical storage from multiple network storage devices into what appears to be a single storage device that is managed from a central console
- Storage virtualization is commonly used in storage area networks (SAN) or (NAS)
- Applications can use storage without having any concern for where it resides, what technical interface it provides, how it has been implemented, which platform it uses and how much of it is available
- **Benefits**
 - Makes the remote storage devices appear local
 - Multiple smaller volumes appear as a single large volume
 - Data is spread over multiple physical disks to improve reliability and performance
 - All operating systems use the same storage device
 - Provided high availability, disaster recovery, improved performance and sharing

virtual storage

→ linux → LVM

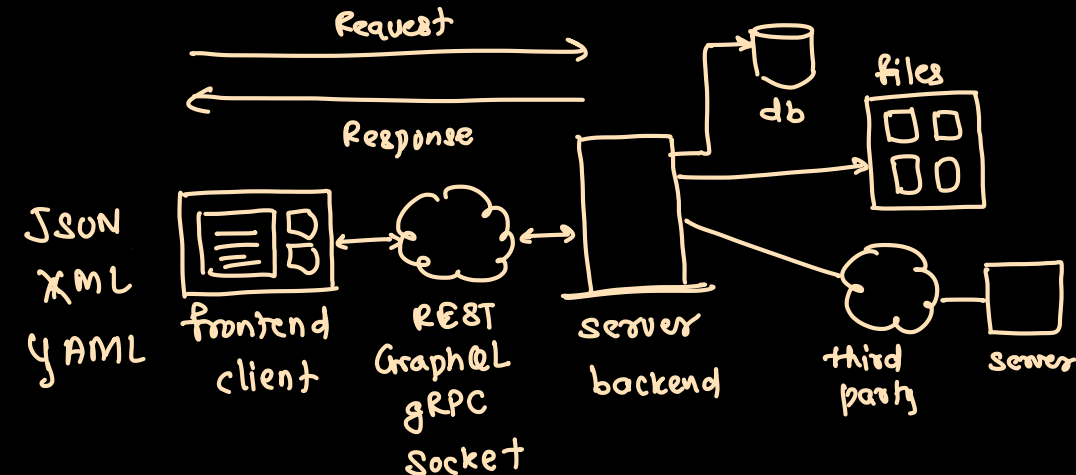
→ windows → RAID
macos / linux



Data virtualization



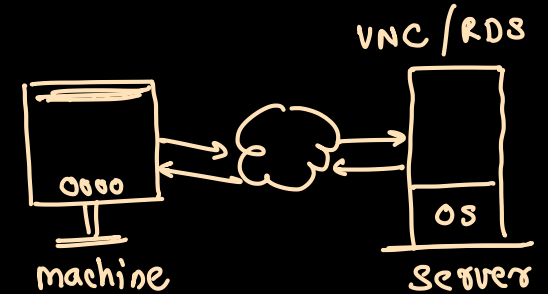
- Data virtualization is the process of aggregating data from different sources of information to develop a single, logical and virtual view of information so that it can be accessed by front-end solutions such as applications, dashboards and portals without having to know the data's exact storage location
- The process of data virtualization involves abstracting, transforming, federating and delivering data from disparate sources
- The main goal of data virtualization technology is to provide a single point of access to the data by aggregating it from a wide range of data sources
- **Benefits**
 - Abstraction of technical aspects of stored data like APIs, Language, Location, Storage structure
 - Provides an ability to connect multiple data sources from a single location
 - Provides an ability to combine the data result sets across multiple sources (also known as data federation)
 - Provides an ability to deliver the data as requested by users



Desktop virtualization



- With desktop virtualization, the goal is to isolate a desktop OS from the endpoint that employees use to access it
- It provides an ability to connect to the desktop from remote site
- When multiple users connect to a shared desktop, as is the case with Microsoft Remote Desktop Services, it's known as shared hosted desktop virtualization
- **Types of Desktop Virtualization**
 - **Virtual Desktop Infrastructure (VDI)**
 - Each user gets a dedicated virtual machine running a desktop OS (e.g., Windows or Linux).
 - Hosted in a centralized server or cloud environment.
 - Provides high customization for each user.
 - **Remote Desktop Services (RDS)**
 - Multiple users share a single operating system instance, usually a Windows Server OS.
 - Resources are more efficiently utilized but offer less customization per user.
 - **Client-Side Virtualization**
 - The virtualized desktop runs locally on the user's device.
 - Examples include using hypervisors like VMware Workstation or Oracle VirtualBox.
 - **Cloud-Based Desktop Virtualization**
 - Also called Desktop as a Service (DaaS).
 - Desktops are hosted and managed in the cloud by a third-party provider.
 - Accessible from any device with an internet connection.



Application Virtualization

→ package of app

- Website → webpack
- android → .apk
- ios → .ipa



- Application virtualization is a technology that encapsulates an application and its dependencies in a way that it can run independently of the underlying operating system or hardware
- This enables the application to operate in an isolated environment, separate from the host system, without the need for direct installation → websites
- **Types of Application Virtualization**
 - **Remote Application Virtualization:**
 - The application is hosted on a remote server and accessed by the user through a client device.
 - Examples: Microsoft RemoteApp, Citrix XenApp.
 - **Streaming Application Virtualization:**
 - The application is delivered to the user's device on-demand. Only the necessary components are downloaded and executed.
 - Example: VMware ThinApp.
 - **Encapsulation-Based Virtualization:**
 - The application is packaged into a portable format and runs directly on the user's device.
 - Example: App-V by Microsoft.

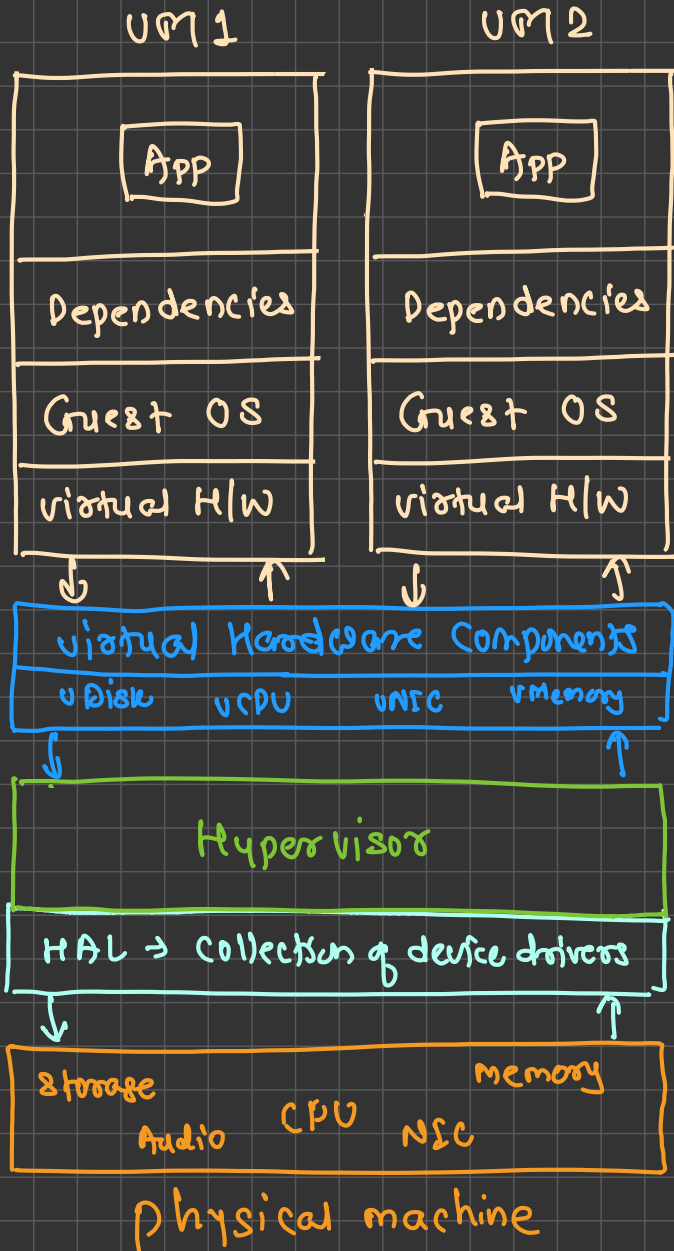
Hardware Virtualization

→ virtual machine

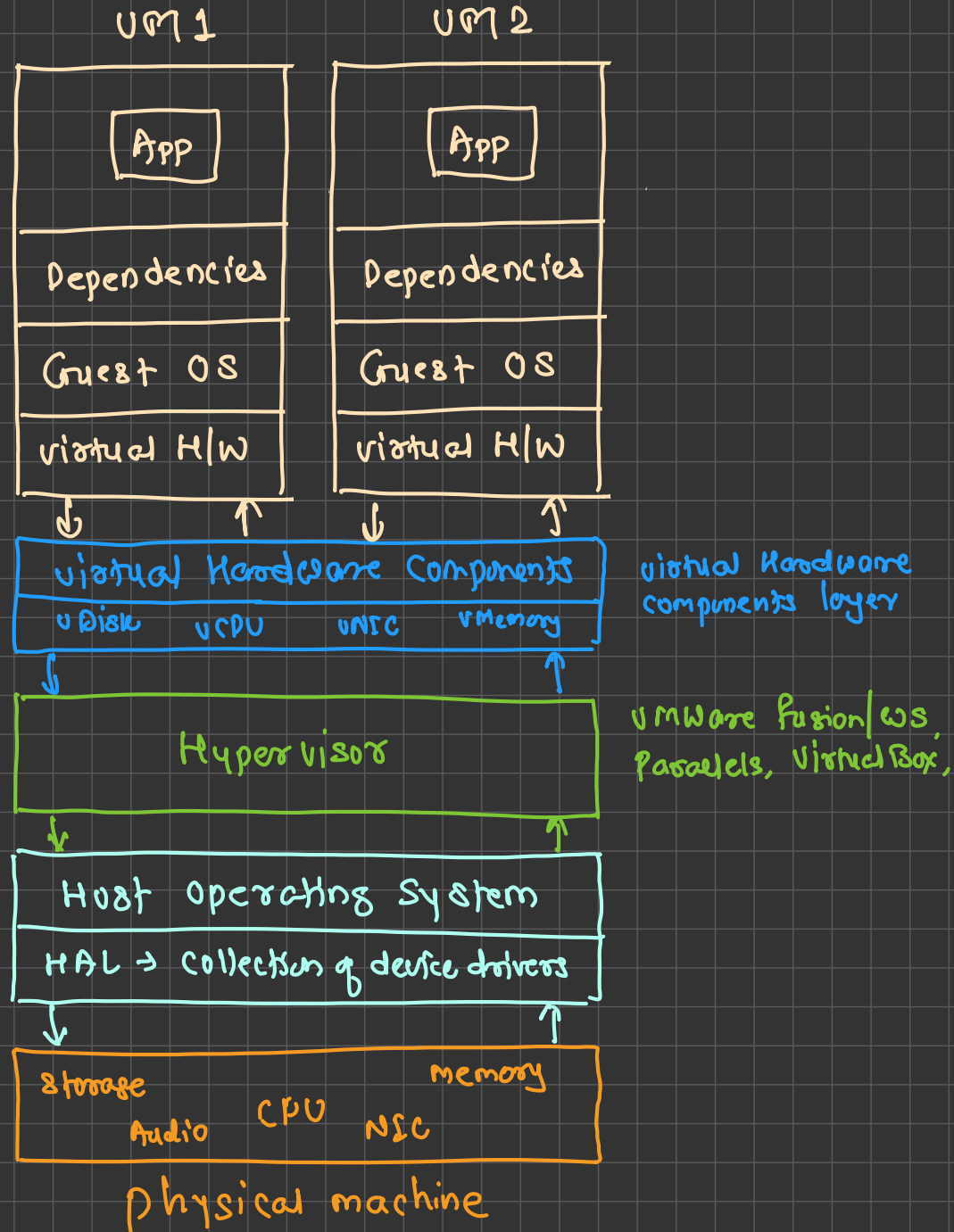
- Hardware virtualization is the process of creating a virtual version of a physical computing environment using a layer of software called a hypervisor
- Instead of running directly on physical hardware, multiple virtual machines (VMs) can share the same physical resources (CPU, memory, storage, network). Each VM behaves like an independent computer with its own operating system and applications
- **Hypervisor (Virtual Machine Monitor – VMM)**
 - The software layer that enables virtualization.
 - Types:
 - Type 1 (Bare-metal): Runs directly on hardware (e.g., VMware ESXi, Microsoft Hyper-V, Xen)
 - Type 2 (Hosted): Runs on top of a host OS (e.g., VirtualBox, VMware Workstation)
- **Virtual Machine (VM)**
 - A software-based emulation of a physical computer
 - Each VM has its own virtual CPU, memory, disk, and network interface
- **Hardware Support**
 - Modern CPUs (Intel VT-x, AMD-V) and hardware (IOMMU, SR-IOV for networking) include extensions to make virtualization efficient and secure

Type I → Bare metal

↳ used by the cloud providers



Type II (Hosted) → users/dev/testers



Types of hardware virtualization



■ Type I

- A Type 1 hypervisor runs directly on the host machine's physical hardware, and it's referred to as a bare-metal hypervisor
- It doesn't have to load an underlying OS first → No Host OS
- With direct access to the underlying hardware and no other software, it is more efficient and provides better performance
- It is best suited for enterprise computing or data centers (cloud providers)
- E.g. VMware ESXi, Microsoft Hyper-V server and open source KVM

■ Type II

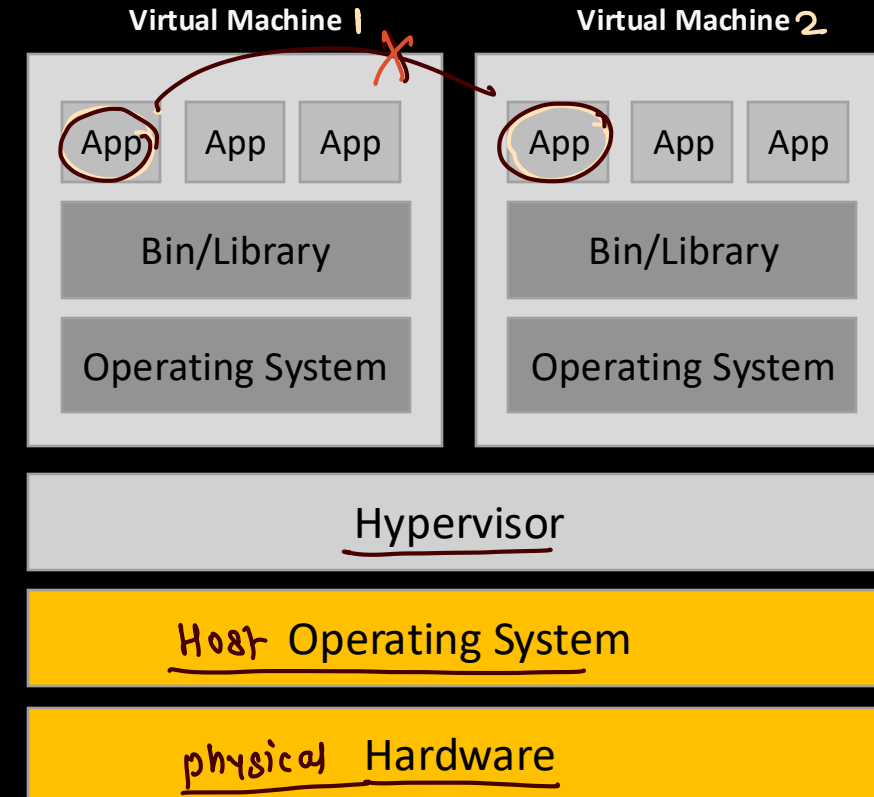
→ Host OS

- A Type 2 hypervisor is typically installed on top of an existing OS, and it's called a hosted hypervisor
- It relies on the host machine's pre-existing OS to manage calls to CPU, memory, storage and network resources
- E.g. VMware Fusion, Oracle VM VirtualBox, Oracle VM Server for x86, Oracle Solaris Zones, Parallels and VMware Workstation

Virtualized Deployment → using virtual machines (type I, type II)



- It allows you to run multiple Virtual Machines (VMs) on a single physical server's CPU
- Virtualization allows applications to be isolated between VMs and provides a level of security as the information of one application cannot be freely accessed by another application
- Virtualization allows better utilization of resources in a physical server and allows better scalability because
 - an application can be added or updated easily
 - reduces hardware costs
- With virtualization you can present a set of physical resources as a cluster of disposable virtual machines
- Each VM is a full machine running all the components, including its own operating system, on top of the virtualized hardware



Advantages of virtualization



- **Lower costs**

- Virtualization reduces the amount of hardware servers necessary within a company and data center
- This lowers the overall cost of buying and maintaining large amounts of hardware

- **Easier disaster recovery**

- Disaster recovery is very simple in a virtualized environment
- Regular snapshots provide up-to-date data, allowing virtual machines to be feasibly backed up and recovered
- Even in an emergency, a virtual machine can be migrated to a new location within minutes

- **Easier testing**

- Testing is less complicated in a virtual environment
- Even if a large mistake is made, the test does not need to stop and go back to the beginning
- It can simply return to the previous snapshot and proceed with the test

- **Quicker backups**

- Backups can be taken of both the virtual server and the virtual machine
- Automatic snapshots are taken throughout the day to guarantee that all data is up-to-date
- Furthermore, the virtual machines can be easily migrated between each other and efficiently redeployed

- **Improved productivity**

- Fewer physical resources results in less time spent managing and maintaining the servers
- Tasks that can take days or weeks in a physical environment can be done in minutes
- This allows staff members to spend majority of their time on more productive tasks, like raising revenue and fostering business initiatives



Cloud

What is cloud computing ?



appliances

- Cloud computing is the delivery of computing resources—like servers, storage, databases, networking, software, and analytics—over the internet (“the cloud”) instead of on local computers or data centers
- Instead of buying and maintaining physical hardware, users can rent and scale resources on demand from a cloud service provider (like AWS, Microsoft Azure, or Google Cloud)
- **Characteristics**
 - On-demand self-service: Users can provision resources whenever needed
 - Broad network access: Services are accessible from anywhere via the internet
 - Resource pooling: Resources (compute, storage, network) are shared among many users
 - Scalability & elasticity: Resources scale up or down automatically as workload changes
 - Pay-as-you-go pricing: You only pay for what you use

↪ cloud providers

CapEx → 0 RS [Data Centers]
OpEx → 100k → 3
 200k → 6
 100k → 3 } Rent

What is Data Center ? → physical infra

physical



- A data center is a facility that houses a large number of computer servers and related IT infrastructure (storage, networking, cooling, power systems) used to store, process, and distribute data and applications
- Think of it as the “factory” where all the computing power and data of organizations or cloud providers live
- **Components of Data Center**
 - Servers – The main computing machines that run applications and store data
 - Storage Systems – Databases, file servers, and storage arrays for keeping information
 - Networking Equipment – Routers, switches, firewalls for connectivity and security
 - Power Supply – Backup generators, UPS (Uninterruptible Power Supply) for 24/7 operation
 - Cooling Systems – Air conditioning, liquid cooling to keep servers from overheating
 - Security Systems – Physical (guards, cameras, biometric access) + Cybersecurity
 - Monitoring Systems – Tools to track server health, performance, and energy usage
- **Types of Data centers** → private data center
 - Enterprise Data Center – Owned and operated by a single organization for its own use
 - Colocation (Colo) – Companies rent space, power, and cooling in a third-party facility
 - Cloud Data Center – Operated by cloud providers (AWS, Azure, Google Cloud) to deliver cloud services
 - Edge Data Center – Smaller facilities located closer to users to reduce latency

Terminologies

scaling $\begin{cases} \text{manual} \\ \text{automatic} \rightarrow \text{autoscaling} \\ \text{Elastic} \end{cases}$

Scalability

- **Definition:** The ability of a system to handle increasing workloads by adding resources (vertical scaling = bigger machine, horizontal scaling = more machines)
- ↳ or decreasing
- **Example:** An e-commerce website adds more servers during a festive sale to handle higher traffic

Elasticity

- **Definition:** The ability of a system to automatically scale resources up or down as needed, usually in real time
- **Example:** A ride-sharing app automatically adds more computing power during rush hours and releases them when demand drops
- **Difference from scalability:** Scalability = capacity to grow, Elasticity = dynamic, automatic adjustment

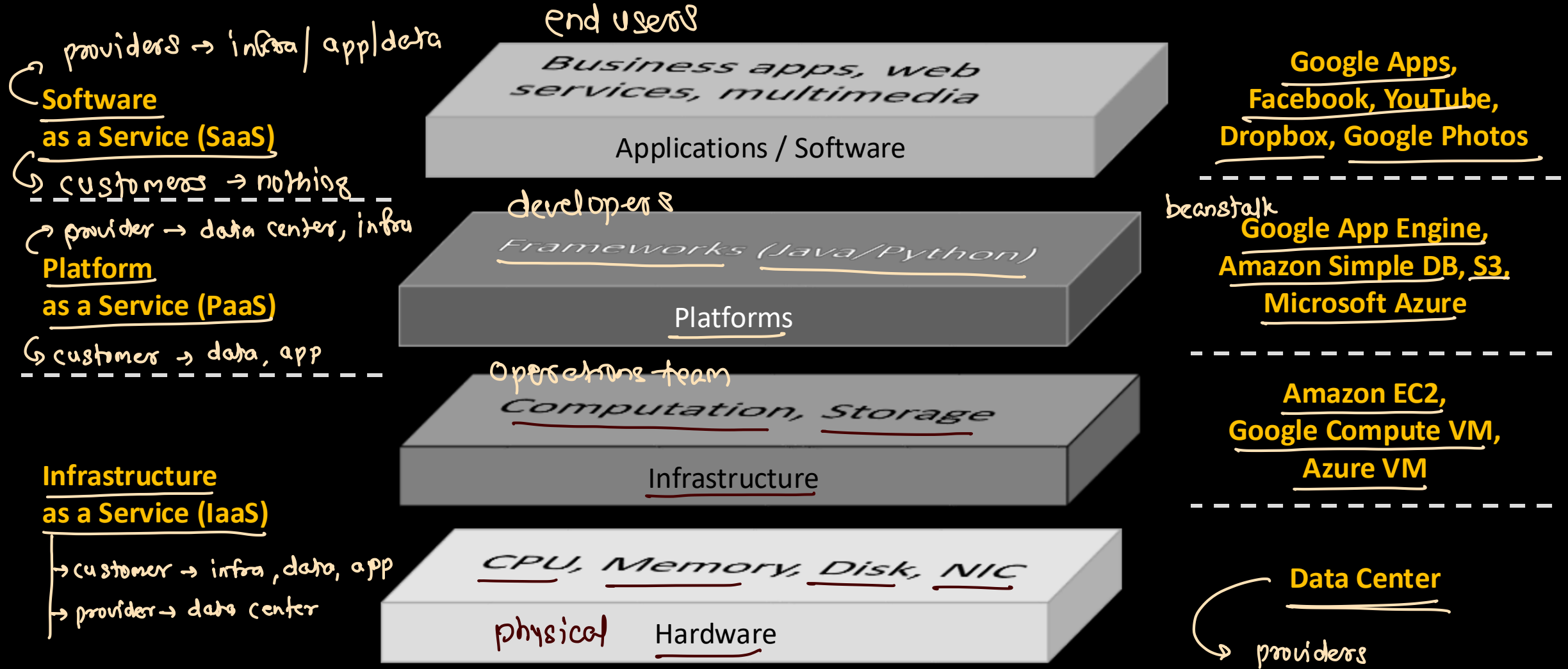
Availability

- **Definition:** The degree to which a system is operational and accessible when needed. Usually expressed as a percentage uptime (e.g., "99.9% availability")
- **Example:** A cloud provider guarantees that your web app will be online and accessible almost all the time (high availability = minimal downtime) → High availability → horizontal scaling (Replication)

On-Demand Service

- **Definition:** A cloud feature that lets users provision and use resources instantly whenever needed without human interaction with the service provider
- **Example:** Launching a new virtual server on AWS with just a few clicks, anytime you want

Service Models



Service Models: IaaS



- Infrastructure as a Service
- Allocates virtualized computing resources to the user through the internet
- IaaS is completely provisioned and managed over the internet
- helps the users to avoid the cost and complexity of purchasing and managing their own physical servers
- Every resource of IaaS is offered as an individual service component and the users only have to use the particular one they need
- The cloud service provider manages the IaaS infrastructure while the users can concentrate on installing, configuring and managing their software
- Generally meant for operations team to setup the required infrastructure
- **Benefits**
 - Time and cost savings: more installation and maintenance of IT hardware in-house,
 - Better flexibility: On-demand hardware resources that can be tailored to your needs,
 - Remote access and resource management.

Service Models: PaaS



- Provides a platform allowing customers to develop, run, and manage applications without the complexity of building and maintaining the infrastructure typically associated with developing and launching an app
- Generally meant for developers
- **Benefits**
 - Mastering the installation and development of software applications
 - Time saving and flexibility for development projects: no need to manage the implementation of the platform, instant production
 - Data security: You control the distribution, protection, and backup of your business data

Service Models: SaaS



- Software as a Service
- Software distribution model in which a third-party provider hosts applications and makes them available to customers over the Internet
- User won't know which computer or operating system or infrastructure is used to host the software
- Generally meant for end user
- **Benefits**
 - You are entirely free from the infrastructure management and aligning software environment: no installation or software maintenance
 - You benefit from automatic updates with the guarantee that all users have the same software version
 - It enables easy and quicker testing of new software solutions.

Cloud Services



- Compute: used to create the Virtual Machine → EC2, lambda (FaaS)
- Storage: used to provide the storage → EBS, EFS, S3
- Database: RDBMS + No SQL → RDS, Redshift, DynamoDB
- Security and Identity Management → IAM
- Media Services →
- Machine Learning → SageMaker, Bedrock, EMR
- Cost Management →
- Application Integration → SNS, SES, SQS

Advantages



- Lower computer costs → No CapEx
- Improved performance → Type I (Base Metal)
- Reduced software costs
- Instant software updates → online
- Improved document format compatibility
- Unlimited storage capacity
- Increased data reliability
- Universal document access
- Latest version availability



Disadvantages

- Requires a constant Internet connection
- Does not work well with low-speed connections
- Features might be limited → depends upon the providers
- Stored data might not be secure → ———
- Stored data can be lost → ———
- Each cloud systems uses different protocols and different APIs

Cloud Providers



- Amazon Web Services
- Google Cloud Platform
- Microsoft Azure
- Rackspace
- DigitalOcean
- Alibaba Cloud
- Oracle Cloud
- IBM Cloud



AWS



What is AWS ?



- AWS (Amazon Web Services) is Amazon's cloud computing platform that provides on-demand IT resources such as servers, storage, databases, networking, AI, DevOps tools, and more — all available over the internet and billed only for what you use
↳ pay as you go
- Before cloud computing:
 - Companies bought physical servers → CapEx
 - Built large data centers → CapEx
 - Spent huge money on maintenance, electricity, cooling, and staff → OpEx
- AWS solved this by giving
 - Virtual servers
 - Global availability
 - Instant scalability
 - Pay-as-you-go pricing

Traditional vs AWS



Feature	Traditional IT	AWS Cloud
✓ <u>Cost</u>	<u>High upfront cost (CAPEX)</u>	<u>Pay-as-you-go (OPEX)</u>
✓ <u>Scalability</u>	<u>Slow, manual</u>	<u>Auto-scaling, instantly</u> → dynamical
✓ <u>Deployment</u>	<u>Days/weeks</u>	<u>Seconds/minutes</u>
✓ <u>Maintenance</u>	<u>You manage hardware</u>	<u>AWS manages hardware</u> (Data Center)
✓ <u>Availability</u>	<u>Needs expensive DR</u>	<u>Multi-region built-in</u> → AZ
✓ <u>Security</u>	<u>On your team</u> – private – more secure	<u>AWS + you (shared)</u> → less secure
✓ <u>Performance</u>	<u>Limited by hardware</u>	<u>Scales globally</u>
✓ <u>Flexibility</u>	<u>Low</u>	<u>Very high</u>
✓ <u>Innovation</u>	<u>Slow</u>	<u>Fast (AI, ML, IoT ready)</u>

AWS Services

→ virtual machines

- **Compute:** Run applications, APIs, backend services, or servers
 - ✓ EC2 – Virtual machines in the cloud → instance → VM
 - ✓ Lambda – Run code without servers (serverless) → FaaS
 - ECS/EKS – Container orchestration
- **Storage:** Store and retrieve files, backups, or large datasets
 - ✓ S3 – Object storage for files → object based storage, EFS → file
 - ✓ EBS – Block storage for EC2 → block based storage, based storage
- **Databases:** Managed databases so you don't maintain servers
 - RDS – MySQL, PostgreSQL, Oracle, etc. → Aurora
 - DynamoDB – NoSQL database → MongoDB compliant, serverless
 - Redshift – Data warehousing
- **Networking:** Connect your applications securely.
 - ✓ VPC – Virtual Private Cloud
 - ✓ Route 53 – DNS services
 - CloudFront – CDN (Content delivery)
- **Security & Identity**
 - ✓ IAM – User permissions
 - KMS – Key management
 - Secrets Manager – Store secrets
- **DevOps & Automation**
 - CodePipeline, CodeBuild, CodeDeploy
 - CloudFormation – Infrastructure as Code
 - CloudWatch – Monitoring → logging
- **Analytics Services**
 - Analyze and process large data.
 - Athena → Query data from S3 using SQL
 - EMR → Big data processing (Hadoop, Spark)
 - Kinesis → Real-time data streaming
 - QuickSight → Business analytics dashboard
- **AI / ML Services**
 - Prebuilt machine learning and model training.
 - SageMaker → Build, train, and deploy ML models
 - Rekognition → Image and video analysis
 - Polly → Text-to-speech
 - Comprehend → Natural language processing (sentiment, extraction)
- **Application Integration**
 - Connect and orchestrate applications.
 - SNS → Push notifications and messages
 - SQS → Message queue service
 - EventBridge → Event-based automation
 - Step Functions → Build workflows

Key Benefits of AWS



- No upfront hardware cost → No CapEx
 - You don't buy servers — you rent them
- Pay only for what you use
 - Like electricity — no usage = no bill
- Highly scalable ↪ Elasticity
 - Scale your app up/down automatically
- Global reach
 - Data centers in 30+ regions worldwide
- High reliability
 - AWS handles hardware failures and redundancy
- Fast deployment
 - Launch servers in seconds instead of weeks

Global Infrastructure: Region

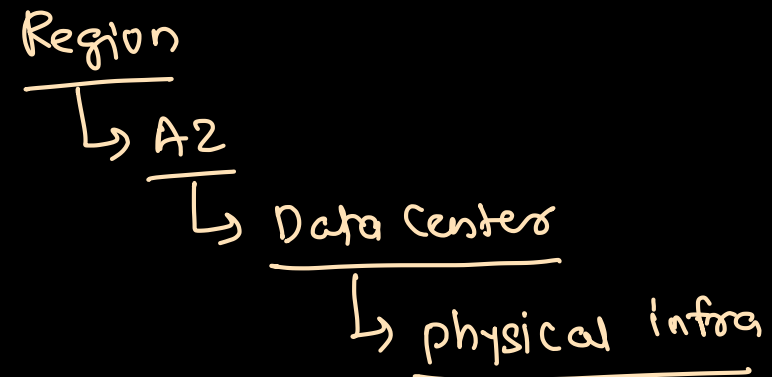
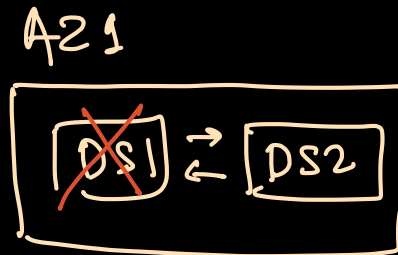


- Geographic area having availability zone(s)
- Collection of availability zones that are geographically located close to one other
- Every Region will act independently of the others, and each will contain at least two Availability Zones
- E.g.
 - US East: N. Virginia, Ohio
 - US West: N. California, Oregon
 - Asia Pacific: Mumbai, Seoul, Singapore, Sydney, Tokyo

Global Infrastructure: Availability Zone



- Essentially the physical data centers of AWS
- This is where the actual compute, storage, network, and database resources are hosted that we as consumers provision within our Virtual Private Clouds (VPCs)
- Availability Zones are always referenced by their Code Name, which is defined by the AZs Region Code Name that the AZ belongs to, followed by a letter
- E.g.
 - the AZs within the eu-west-1 region (EU Ireland), are
 - eu-west-1a
 - eu-west-1b
 - eu-west-1c



Global Infrastructure: Edge Locations



- Edge Locations are AWS sites deployed in major cities and highly populated areas across the globe
- Generally used to cache data and reduce latency for end-user access by using the Edge Locations as a global Content Delivery Network (CDN)
- Edge Locations are primarily used by end users who are accessing and using your services
- E.g.
 - Route 53: DNS Lookup
 - CloudFront
 - Content Delivery Network (CDN)
 - Cached contents, streaming distribution, acceleration



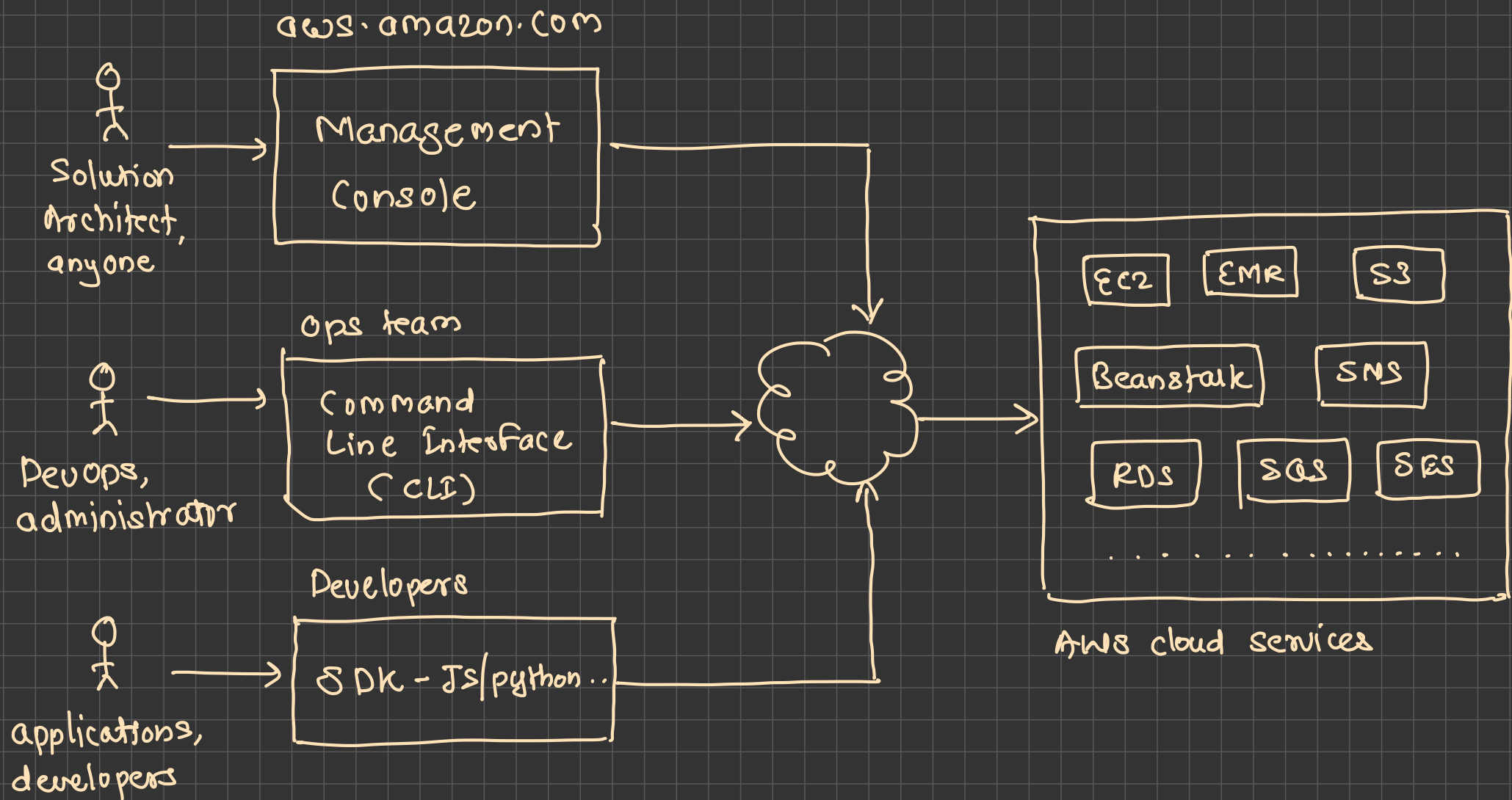
EC2

EC2



↪ virtual machine

- Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides resizable **compute capacity** in the cloud
- It is a virtual machine you will be building in the cloud
- EC2 instances are designed to mimic traditional on-premise servers, but with the ability to be commissioned and decommissioned on-demand for easy scalability and elasticity
- EC2 supports variety of operating systems: → AMI
 - Linux: Amazon Linux, Ubuntu, Red Hat Enterprise, SUSE Linux Enterprise Server, Fedora, Debian, CentOS, Gentoo Linux, Oracle Linux, FreeBSD
 - Windows: Windows Server, Windows , macOS
- Every instance comprised of
 - Amazon Machine Image (AMI) → OS disk image
 - Instance type → CPU / RAM / GPU
 - Network Interface (s)
 - Storage → Disk (EBS)
 - security group → firewall



Amazon Machine Image (AMI) → region specific



- Operating System used to create virtual machine (EC2 instance)
- AMI are built for a specific region
- You can copy an AMI from one region to another → autoscaling
- You can also create a custom AMI with required applications/configuration
- AMI contains
 - Template for root volume → OS installation
 - Launch permissions that control which account can use the AMI
 - EBS mapping that specifies the volume(s) to attach the instance when its launched → storage
- AMI comes into two types
 - Instance store backed AMI
 - EBS backed AMI ↵

Instance Type



- Used to decide the EC2 instance configuration
- AWS provides various instance types [<https://aws.amazon.com/ec2/instance-types/>]
 - General purpose: A (ARM), T (Cheapest), M (Main)
 - Compute optimized: C (Compute)
 - Memory optimized: R (RAM), X (Extreme RAM), Z (High compute and memory)
 - Accelerated computing: P (Picture-GPU), G (Graphics), F (Fast)
 - Storage optimized: I (IOPS), D (Data), H (High Disk Throughput)

Instance Types



Type	Category	Description	Use Cases
M5	General Purpose	Balance of compute, memory and network resources	Mid-sized databases
C5	Compute Optimized	Advanced CPUs	Modelling, Analytics
H1	Storage Optimized	Local HDD Storage	Map Reduce
R4	Memory Optimized	More RAM for \$	In-memory caching
X1	Memory Optimized	Terabytes of RAM and SSD	In-memory database
I3	IO Optimized	Local SSD storage, high IOPS	NoSQL databases
G3	GPU Graphics	GPUs with video encoders	3d rendering
P3	GPU Compute	GPUs with tensor cores	Machine Learning
F1	Accelerated Computing	FPGA, custom hardware accelerations	Genomics
T2	Burstable	Shared CPUs, lowest cost	Web servers

Security Group



- Acts as a virtual firewall for your instance to control inbound and outbound traffic
- Controls the ports and protocols that can reach the front-end listener
- Every EC2 instance must have at least one security group attached
- Up to 5 security groups can be attached to an EC2 instance
- Security groups act at the instance level, not the subnet level
- Security group contains rules
 - You can specify allow rules, but not deny rules
 - You can specify separate rules for inbound and outbound traffic
 - When you create a security group, it has no inbound rules
 - By default, a security group includes an outbound rule that allows all outbound traffic
 - Security groups are stateful
 - Instances associated with a security group can't talk to each other unless you add rules allowing it
 - Security groups are associated with network interfaces

EC2 Key Pairs



- Uses PEM format (Privacy Enhanced Mail)
- Used to authenticate a client when logging into EC2 instance
- Each key pair consists of a public key and a private key
- AWS stores the public key on the instance and you are responsible for storing the private key
- To log into the instance you must create and authenticate with key pair
 - Linux instances have no password and you use a key pair to log in
 - With windows you use a key pair to obtain the administrator password and then log into the instance with RDP
- During the creation process of an EC2 instance you are required to either create a new key pair or use existing pair
- The private key is available for download and stored on your local drive
- NOTE: it will be available only once in the form of .pem file